

Aman Kumar

Software Development Engineer | IIT Roorkee Graduate '26

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Professional Summary

IIT Roorkee graduate (B.Tech + M.Tech, 2026) with hands-on experience designing, developing, testing, and deploying scalable, maintainable software applications using Python, JavaScript, and C++. Built full-stack applications with FastAPI/Flask REST backends, React + Redux frontends, normalised relational database schemas, and LangGraph AI agents — all following clean modular architecture, Pydantic validation, and ORM-backed data access. Internship experience at an AI startup with Agile sprint cycles and structured code reviews. Competitive programmer: Codeforces 1550 (Specialist), CodeChef 2100 (5-Star), 1500+ problems solved — rigorous analytical problem-solving applied directly to software design and debugging.

Education

Year	Degree / Examination	Institution / Board	CGPA / %
2026	B.Tech + M.Tech — Geophysical Technology	Indian Institute of Technology, Roorkee	6.5
2021	Intermediate (Class XII)	Gaya College (BSEB), Gaya, Bihar	77.40%
2019	Matriculate (Class X)	B.L Indo Anglian Public School (CBSE)	86.80%

Technical Skills

Languages	Python, JavaScript, C++ — all OOP; proficient in clean, efficient, maintainable code
Backend & APIs	FastAPI, Flask, REST API Design, JWT Authentication, SQLAlchemy ORM, Pydantic, Git, GitHub, Linux
Frontend	React, Redux (state management), HTML5, CSS3, JavaScript, Jinja2, Responsive Web Design
Databases	PostgreSQL, MySQL, SQLite, Neo4j, Firebase Firestore
Testing & Process	Unit Testing, Integration Testing, Code Review, Debugging, Agile/Scrum, Technical Documentation
AI/ML (Plus)	LangChain, LangGraph, HuggingFace, FAISS, RAG Pipelines, OpenAI/Groq API, Vector Embeddings

Work Experience

Software Development Intern | CallSure AI

Jun 2025 – Jul 2025

- Designed, developed, tested, and deployed a production Flask REST API backend for a customer support platform — ticket ingestion, priority routing, and resolution retrieval with a normalised MySQL schema across 5+ categories, applying clean code, secure coding, and separation-of-concerns throughout.
- Implemented an automated query classification module with unit-tested logic across 3+ urgency tiers; participated in code reviews with the senior developer at every merge to enforce quality standards and consistent coding practices in an Agile sprint environment.
- Integrated an AI-powered retrieval layer (LangChain + FAISS) as a backend service, demonstrating ability to develop and integrate REST APIs and external AI/ML tools into production applications.

Projects

AI-First CRM — HCP Module | FastAPI, LangGraph, React, Redux, PostgreSQL, Groq

Jun 2025 – Jul 2025

- Built a full-stack AI CRM for healthcare professionals: FastAPI backend with CRUD REST routes and a LangGraph AI agent (Groq LLM) — dual input modes (structured form + conversational AI chat) both persisting to the same PostgreSQL database via SQLAlchemy ORM.
- Implemented a 5-tool LangGraph agent covering the end-to-end sales rep workflow — Pydantic-validated inputs, error-safe DB writes, and typed state transitions across 3 normalised PostgreSQL tables.

LostFound — Full-Stack Web Application | Python, Flask, SQLite, JavaScript, HTML5/CSS3

Oct 2024 – Nov 2024

- Designed, developed, and deployed a full-stack web application using Python/Flask (backend) and HTML5/CSS3/JavaScript (frontend) serving 5000+ IIT Roorkee students — RESTful API routes, normalised SQLite schema with ORM, secure session management, and responsive UI components.
- Applied unit and integration testing at each feature milestone; conducted code reviews at every merge enforcing quality, security, and standards; maintained technical documentation on GitHub.

Document Retrieval System | Python, LightRAG, Neo4j, FAISS, FastAPI

Jan 2025 – Mar 2025

- Architected a scalable Python backend system with dual-level indexing (Neo4j graph + FAISS vector store), optimised query routing, and data integrity across 50+ page document corpora — achieving ~65% reduction in processing cost over existing approaches through careful performance and scalability tuning.
- FastAPI endpoints for document upload and query processing; SQLAlchemy-backed metadata; maintained clean, well-documented codebase on GitHub with architecture documentation and reproducible setup steps.

Achievements

- Attained Codeforces rating of 1550 (Specialist) and CodeChef rating of 2100 (5-Star); solved 1500+ algorithmic problems — strong OOP, data structures, and analytical problem-solving fundamentals.
- Delivered full working software applications within 24-hour hackathon constraints at IIT Roorkee, balancing speed, code quality, and reliability under pressure.
- Secured top 1% rank in JEE Mains and JEE Advanced among 1.2M+ candidates; admitted to IIT Roorkee — one of India's premier engineering institutes.

Hobbies & Interests

Football & Cricket: Enjoy playing and following both sports; regular viewer of international tournaments. **Travelling:** Like exploring new places and experiencing different cultures. **Gaming:** Regular PUBG player — team strategy and quick decision-making under pressure. **Tech Trends:** Stay updated with emerging technologies, software engineering best practices, and industry developments through blogs, formal courses, and self-directed learning.